Cosmological solutions with torsion effects in Polynomial Affine Gravity

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1 Introduction

Einstein's theory of General Relativity, it is a gravitational theory where the interaction is solely mediated by the metric tensor [1, 2]. Providing an ansatz compatible with the symmetries of the observable universe (cosmological principle), it is possible to model the Universe with Einstein's field equations [3].

With the introduction of the cosmological constant (see Ref. [4]) it is possible to obtain nontrivial dynamics to describe the evolution of the universe, leading to nine different scenarios, depending on the values of the geometrical factor κ (which characterize the geometry of the spatial foliation of the Universe), and the cosmological constant Λ. Although, the introduction of a cosmological constant does indeed introduce a repulsive force required to keep the universe in static equilibrium, this idea does not provide a precise description of the current state of the universe. In order to have a more accurate description of the observable universe, it is required the introduction of the energy-momentum tensor, along with the state equation. By taking into account this new information into Einstein's Field Equations, it is possible to portray the observable universe. Moreover, it allow us to clearly distinguish three different epochs, *evolving* from a radiation era, to a matter era and finally to an accelerated expansion due to dark energy presence. This is the universe describe by Friedmann–Lemaitre–Robertson–Walker (FLRW), see Refs. [5–11], and provides the foundations for the standard model of cosmology ΛCDM.

Einstein's formulation its the best theory we have so far to describe gravitational interactions, and there is an agreement between the theoretical predictions with the observational data, to see a full review on its empirical evidence, see Ref. [12,13]. Perhaps, one of its more important theoretical predictions, it is the existence of gravitational waves, which were detected by by LIGO-Virgo collaborations [14, 15]. Despite its success, there are still nontrivial problems related to its description of the geometry of space-time, specifically problems related to a quantization of gravity [16–19] and the necessity of the dark sector to explain the galaxy rotation curve speed [20, 21], along with accelerated expansion [22].

These type of problems suggest, in principle, Einstein's theory of gravity must be an effective theory of another gravitational theory. In that sense, there

are extensions of general relativity such as where the Einstein-Hillbert action its replaced by an arbitrary function, named $f(\mathcal{R})$, which was first proposed in Ref. [23]. An interesting case of this type of gravity, its the Starobinsky's model [24] that allows to describe a universe without a singularity. Moreover, due to quantum corrections to general relativity, lead to a new contribution to the Einstein-Hillbert action due to curvature squared new term [25], which is a special case of $f(\mathcal{R})$ gravity.

Because the choice on function $f(\mathcal{R})$ it is an arbitrary one, there is an abundance of cosmological models, however, one could go even further and at a fundamental level, claim, that the metric tensor and the affine connection (a prior) must be independent fields, these are the type of modify models are known as *metric-affine* models, and its formalism it is known as *Palatini* formalism. There is an extensive collection on the literature related to metric-affine models [26–30], and an extensive/complete review can be found in Ref. [31].

In addition, its possible to built models of gravity with the affine connection as its fundamental field. The first affine model of gravity was proposed by Sir Eddington [32], and pure affine formalism were proposed in Refs. [33–35]. These type of affine formulations models of gravity, use the square-root of the Ricci tensor instead of the metric tensor to define the invariant volume element

There are other extensions to Einstein's theory of general relativity, for example Einstein-Cartan model [36,37], which uses the Einstein-Hillbert action to define gravitational interactions, with a non-symmetric connection by introducing nontrivial torsion effects. Moreover, the inclusion of extra dimensions where first proposed and studied by T. Kaluza and O. Klein in Refs. [38], specifically, uses a fifth dimension which allows to unify the gravitational with an electromagnetism forces. There are many gravitational models and within different classifications, an illustrative example of a so-called *map of modify gravity* can be found in Ref. [39].

In order to built a purely affine gravitational model of gravity, the metric tensor can not play any role in its formulation and in its building blocks, and this is where the Polynomial Affine Gravity enters. The Polynomial Affine Gravity its an alternative and purely affine model of gravity whose fundamental field is the affine connection instead of the metric tensor. The model has been studied in Refs. [40–46] and on this work, we explore the consequences of including torsion effects in its field equations in four-dimensions.

This paper is organized as follows: In Section 2, we present a brief overview on how to built the Polynomial Affine Gravity model, highlighting its features, and how it is possible to still have two different emergent-metrics structures despite being a purely affine model. A complete scan of cosmological solutions to the field equations is presented in Section 3. In Section 4, we analyse and discuss the cosmological solutions. Final remarks are presented in Section 5.

2 Polynomial Affine Gravity

The Polynomial Affine Gravity is a model of gravity whose fundamental field is the affine connection $\hat{\Gamma}$,[1] whose irreducible fields can be decomposed into its symmetric and skew symmetric part

$$\begin{aligned}\hat{\Gamma}_\alpha{}^\beta{}_\gamma &= \hat{\Gamma}_{(\alpha}{}^\beta{}_{\gamma)} + \hat{\Gamma}_{[\alpha}{}^\beta{}_{\gamma]}, \\ &= \Gamma_\alpha{}^\beta{}_\gamma + \mathcal{B}_\alpha{}^\beta{}_\gamma + \delta^\beta_{[\gamma}\mathcal{A}_{\alpha]},\end{aligned} \tag{1}$$

the first term corresponds to the symmetric part of the connection $\hat{\Gamma}_{(\alpha}{}^\beta{}_{\gamma)} = \Gamma_\alpha{}^\beta{}_\gamma$, whereas the last two terms are related to the torsion tensor, the former traceless component of the torsion $\mathcal{B}_\alpha{}^\beta{}_\gamma$ and the latter corresponds to its vectorial part $\mathcal{A}_\alpha$.

Since the symmetric part of the connection does not preserve the invariance under diffeomorphisms, it enters into the action through the covariant derivative ∇^Γ. Additionally, the volume form is defined by the wedge product (which does not required the metric tensor to be well-defined) $\mathrm{d}V^{\alpha\beta\gamma\delta} = \mathrm{d}x^\alpha \wedge \mathrm{d}x^\beta \wedge \mathrm{d}x^\gamma \wedge \mathrm{d}x^\delta$.

Therefore, the fundamental building blocks in our affine model of gravity are ∇, $\mathcal{B}$, $\mathcal{A}$ and $\mathrm{d}V$. The action is built up using a sort of *dimensional analysis* technique. The method allows us to consider every scalar densities composed by powers of the irreducible fields of the affine connection [41, 44]. In the same spirit the three-dimensional model was also built up Ref. [45].

The most general action (up to topological invariants and boundary terms) in four dimensions is given by

$$\begin{aligned}S = \int \mathrm{d}V^{\alpha\beta\gamma\delta}\Big[&B_1\mathcal{R}_{\mu\nu}{}^\mu{}_\rho\mathcal{B}_\alpha{}^\nu{}_\beta\mathcal{B}_\gamma{}^\rho{}_\delta + B_2\mathcal{R}_{\alpha\beta}{}^\mu{}_\rho\mathcal{B}_\gamma{}^\nu{}_\delta\mathcal{B}_\mu{}^\rho{}_\nu + B_3\mathcal{R}_{\mu\nu}{}^\mu{}_\alpha\mathcal{B}_\beta{}^\nu{}_\gamma\mathcal{A}_\delta \\ &+ B_4\mathcal{R}_{\alpha\beta}{}^\sigma{}_\rho\mathcal{B}_\gamma{}^\rho{}_\delta\mathcal{A}_\sigma + B_5\mathcal{R}_{\alpha\beta}{}^\rho{}_\rho\mathcal{B}_\gamma{}^\sigma{}_\delta\mathcal{A}_\sigma + C_1\mathcal{R}_{\mu\alpha}{}^\mu{}_\nu\nabla_\beta\mathcal{B}_\gamma{}^\nu{}_\delta \\ &+ C_2\mathcal{R}_{\alpha\beta}{}^\rho{}_\rho\nabla_\sigma\mathcal{B}_\gamma{}^\sigma{}_\delta + D_1\mathcal{B}_\nu{}^\mu{}_\lambda\mathcal{B}_\mu{}^\nu{}_\alpha\nabla_\beta\mathcal{R}_\gamma{}^\lambda{}_\delta + D_2\mathcal{B}_\alpha{}^\mu{}_\beta\mathcal{B}_\mu{}^\lambda{}_\nu\nabla_\lambda\mathcal{B}_\gamma{}^\nu{}_\delta \\ &+ D_3\mathcal{B}_\alpha{}^\mu{}_\nu\mathcal{B}_\beta{}^\lambda{}_\gamma\nabla_\lambda\mathcal{B}_\mu{}^\nu{}_\delta + D_4\mathcal{B}_\alpha{}^\lambda{}_\beta\mathcal{B}_\gamma{}^\sigma{}_\delta\nabla_\lambda\mathcal{A}_\sigma + D_5\mathcal{B}_\alpha{}^\lambda{}_\beta\mathcal{A}_\sigma\nabla_\lambda\mathcal{B}_\gamma{}^\sigma{}_\delta \\ &+ D_6\mathcal{B}_\alpha{}^\lambda{}_\beta\mathcal{A}_\gamma\nabla_\lambda A_\delta + D_7\mathcal{B}_\alpha{}^\lambda{}_\beta\mathcal{A}_\lambda\nabla_\gamma A_\delta + E_1\nabla_\rho\mathcal{B}_\alpha{}^\rho{}_\beta\nabla_\sigma\mathcal{B}_\gamma{}^\sigma{}_\delta \\ &+ E_2\nabla_\rho\mathcal{B}_\alpha{}^\rho{}_\beta\nabla_\gamma\mathcal{A}_\delta + F_1\mathcal{B}_\alpha{}^\mu{}_\beta\mathcal{B}_\gamma{}^\sigma{}_\delta\mathcal{B}_\mu{}^\lambda{}_\rho\mathcal{B}_\sigma{}^\rho{}_\lambda + F_2\mathcal{B}_\alpha{}^\mu{}_\beta\mathcal{B}_\gamma{}^\nu{}_\lambda\mathcal{B}_\delta{}^\lambda{}_\rho\mathcal{B}_\mu{}^\rho{}_\nu \\ &+ F_3\mathcal{B}_\nu{}^\mu{}_\lambda\mathcal{B}_\mu{}^\nu{}_\alpha\mathcal{B}_\beta{}^\lambda{}_\gamma\mathcal{A}_\delta + F_4\mathcal{B}_\alpha{}^\mu{}_\beta\mathcal{B}_\gamma{}^\nu{}_\delta\mathcal{A}_\mu\mathcal{A}_\nu\Big].\end{aligned} \tag{2}$$

In the above action, the covariant derivative and the curvature tensor are defined with respect to the symmetric part of the connection, meaning $\nabla = \nabla^\Gamma$ and $\mathcal{R} = \mathcal{R}^\Gamma$.

One of the most important features of this model is that lack of the metric tensor in its formulation, causes that the number of terms in the action is finite, we called this property *rigidity*.

[1] The action is polynomial in the connection and its derivative

Additionally, even though, the manifold is endowed with an affine structure $\hat{\Gamma}$, it is still possible to find emergent (connection descendent) metrics in the space of solutions. The first one comes from the Ricci tensor $\mathcal{R}_{\mu\nu}$, which is defined by the contraction of the Riemann tensor.[2]

$$\hat{\Gamma} \to \nabla^{\Gamma} \to \mathcal{R}_{\alpha\beta}{}^{\gamma}{}_{\delta} \to \mathcal{R}_{\beta\delta}. \tag{3}$$

A second metric structure, comes from the contraction of the product of two torsion tensor.[3] This idea was first introduce by Poplwaski in Ref. [47], and the metric structure its defined as follow

$$\mathcal{P}_{\alpha\delta} = \left(\mathcal{B}_{\alpha}{}^{\beta}{}_{\gamma} + \delta^{\beta}_{[\gamma}\mathcal{A}_{\alpha]}\right)\left(\mathcal{B}_{\beta}{}^{\gamma}{}_{\delta} + \delta^{\gamma}_{[\delta}\mathcal{A}_{\beta]}\right). \tag{4}$$

Interestingly, all the coupling constant are dimensionless, which suggest that the model is power-counting renormalizable. This, is a necessary condition but no sufficient condition to guarantee that the Polynomial Affine Gravity is renormalizable. Additionally, it suggest a conformal symmetry, at least at a classical level.

Moreover, in the hypothetical scenario of its quantization, because of its *rigidity*, all possible counter-terms will have the form of the ones already written in eq.(2). This is something desirable from a stand point of Quantum Field Theory view, for the reason that the superficial degree of divergence vanishes.

Finally, in the torsion-free sector it is possible to recover all Einstein's vacuum solutions. Even more, it is possible to coupled a scalar field using the *dimensional analysis* technique with the torsion tensor and the volume form, while the potential energy re-scale the volume element. From which, there is a subspace of solutions that allow us to recover Einsteins's field equations coupled with a scalar field, see Ref.[Paper˙Inflacion]

Since we are interested in the study of the cosmology, we want to impose the symmetries of the cosmological principle in our fundamental field $\hat{\Gamma}$ and in its irreducible fields. In order to build up the ansatz we compute the Lie derivative of each irreducible field along the Killing vectors ξ_i that generates the symmetries of homogeneity (translations) $\mathcal{P}_i$ and isotropy (rotations) $\mathcal{J}_i$. An explicit computation of the Lie derivative can be found in Ref. [48]. In what follows we shall show only the results.

The computation of the Lie derivative of Γ along the Killing vectors, determined its coefficients:

$$\Gamma_t{}^t{}_t = f(t), \qquad \Gamma_i{}^t{}_j = g(t)S_{ij}, \tag{5}$$

$$\Gamma_t{}^i{}_j = h(t), \delta^i_j \qquad \Gamma_i{}^j{}_k = \gamma_i{}^j{}_k, \tag{6}$$

where S_{ij} is the three-dimensional rank two symmetric tensor defined as follow

$$S_{ij} = \begin{pmatrix} \frac{1}{1-\kappa r^2} & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2\sin^2\theta \end{pmatrix}, \tag{7}$$

[2]The Riemann tensor its defined by the commutator of covariant derivatives acting on a vector, therefore, it does not required the existence of a metric structure.

[3]This comes from the antisymmetric part of the affine connection

and γ is the symmetric connection compatible with desired symmetries written as

$$\gamma_r{}^r{}_r = \frac{\kappa r}{1-\kappa r^2}, \qquad \gamma_\theta{}^r{}_\theta = \kappa r^3 - r, \quad \gamma_\varphi{}^r{}_\varphi = \left(\kappa r^3 - r\right)\sin^2\theta, \quad \gamma_r{}^\theta{}_\theta = \frac{1}{r}, \tag{8}$$

$$\gamma_\varphi{}^\theta{}_\varphi = -\cos\theta\sin\theta, \quad \gamma_r{}^\varphi{}_\varphi = \frac{1}{r}, \qquad \gamma_\theta{}^\varphi{}_\varphi = \frac{\cos\theta}{\sin\theta}. \tag{9}$$

Additionally, the affine function $f(t)$ can be set equal to zero, under a parametrization of the time coordinate, for more information on this type of transformation, refer to Ref. [45]. Therefore, there are only two non trivial functions to define completely the symmetric part of the connection.

Following a similar procedure for the torsion tensor, is possible to determined the ansatz compatible with the required symmetries. For its traceless part $\mathcal{B}$, the non-vanishing components are

$$\mathcal{B}_\theta{}^r{}_\varphi = \psi(t) r^2 \sin\theta\sqrt{1-\kappa r^2}, \quad \mathcal{B}_r{}^\theta{}_\varphi = \frac{\psi(t)\sin\theta}{\sqrt{1-\kappa r^2}}, \quad \mathcal{B}_r{}^\varphi{}_\theta = \frac{\psi(t)}{\sqrt{1-\kappa r^2}\sin\theta}, \tag{10}$$

while the nontrivial components of its vectorial part $\mathcal{A}$ is

$$\mathcal{A}_t = \eta(t). \tag{11}$$

Using the cosmological ansatz presented in eq. (5), the Ricci tensor is defined as follow

$$\mathcal{R}_{tt} = \dot{h} + h^2 \qquad \mathcal{R}_{rr} = \frac{\dot{g} + gh + 2\kappa}{1-\kappa r^2}, \tag{12}$$

moreover, the Poplawski's metric is computed using eqs. (10) and (11)

$$\mathcal{P}_{tt} = \eta^2 \qquad \mathcal{P}_{rr} = -\frac{2\psi^2}{1-\kappa r^2}. \tag{13}$$

In order to use the Ricci tensor $\mathcal{R}_{\mu\nu}$ or the Poplawski $\mathcal{P}_{\mu\nu}$ as an emergent metric tensor, is necessary that the tensor is well defined, meaning, it can be invertible. If the tensors are well behaved, then the factors $(\dot{g} + gh + 2\kappa)$ or $\left(-2\psi^2\right)$ play an analogue role to the scale factor $a^2(t)$ from FLRW.

Finally, the field equations are obtained by varying the action with respect to each irreducible field using Kijowkski's formalism, see Ref. [49], which we also reviewed in Ref. [44]. The complete set of the field equation for each irreducible field can be found in Ref. [50].

3 Cosmological solutions

Using the ansatz found in eqs. (5), (10) and (11) the field equation are

$$\left(B_3\left(\dot{g} + gh + 2\kappa\right) - 2B_4\left(\dot{g} - gh\right) + 2D_6\eta g - 2F_3\psi^2\right)\psi = 0, \tag{14}$$

$$\left(B_3\eta\psi - 2B_4\eta\psi + C_1\left(\dot{\psi} - 2h\psi\right)\right)g = 0, \tag{15}$$

$$\left(B_3 + 2B_4\right)\eta g\psi + 2C_1\left(\kappa\psi + 4gh\psi - g\dot{\psi} - \psi\dot{g}\right) + 2\psi^3\left(2D_2 - D_1 - D_3\right) = 0, \tag{16}$$

$$\begin{aligned} &B_3\left(\eta\left(h\psi - \dot{\psi}\right) - \psi\dot{\eta}\right) - 2B_4\left(\eta\left(-h\psi - \dot{\psi}\right) - \psi\dot{\eta}\right) \\ &+ C_1\left(4h^2\psi + 2\psi\dot{h} - \ddot{\psi}\right) + D_6\eta^2\psi = 0, \end{aligned} \tag{17}$$

$$\begin{aligned} &B_3\left(\dot{g} + gh + 2\kappa\right)\eta - 2B_4\left(\dot{g} - gh\right)\eta + C_1\left(2\kappa h + 4gh^2 + 2g\dot{h} - \ddot{g}\right) \\ &+ 6h\psi^2\left(2D_2 - D_1 - D_3\right) + D_6\eta^2 g - 6F_3\eta\psi^2 = 0 \end{aligned} \tag{18}$$

where $\mathcal{A}$ and $\mathcal{B}$ have only one equation, eq. (14) and eq. (18) respectively, whereas there are three field equations for Γ, which are eqs.(15) to (17). Notice, we have four unknown functions of time $g(t)$, $h(t)$, $\psi(t)$ and $\eta(t)$ while there are five differential equations, meaning the system is overdetermined.

Nonetheless, we shall show that, it can actually be solved analytically without any assumption. For this end, we develop a *logical scheme* that allow us to seek (systematically) for solutions. First, notice eqs. (14) and (15) are of the form

$$\mathcal{F}(g, \dot{g}, h, \psi, \eta)\psi = 0, \tag{19}$$

$$\mathcal{G}(h, \psi, \dot{\psi}, \eta)g = 0, \tag{20}$$

where

$$\mathcal{F}(g, \dot{g}, h, \psi, \eta) = B_3\left(\dot{g} + gh + 2\kappa\right) - 2B_4\left(\dot{g} - gh\right) + 2D_6\eta g - 2F_3\psi^2, \tag{21}$$

$$\mathcal{G}(h, \psi, \dot{\psi}, \eta) = B_3\eta\psi - 2B_4\eta\psi + C_1\left(\dot{\psi} - 2h\psi\right). \tag{22}$$

Thus, from eqs. (19) and (20) its possible to distinguish four different branches:

$$\textbf{First branch:} \quad \mathcal{F}(g, h, \psi, \eta) = 0 \quad \wedge \quad \mathcal{G}(h, \psi, \eta) = 0, \tag{23}$$

$$\textbf{Second branch:} \quad \mathcal{F}(g, h, \psi, \eta) = 0 \quad \wedge \quad g = 0, \tag{24}$$

$$\textbf{Third branch:} \quad \mathcal{G}(h, \psi, \eta) = 0 \quad \wedge \quad \psi = 0, \tag{25}$$

$$\textbf{Fourth branch:} \quad \psi = 0 \quad \wedge \quad g = 0. \tag{26}$$

In terms of the richness of the solutions, the most interesting branch is that from eq. (23), which is the most general case, while the simplest is the one characterized by eq. (26).

3.1 First branch

This branch has the most general case, where the field equations are

$$B_3\left(\dot{g}+gh+2\kappa\right)-2B_4\left(\dot{g}-gh\right)+2D_6\eta g-2F_3\psi^2=0, \tag{27}$$

$$B_3\eta\psi-2B_4\eta\psi+C_1\left(\dot{\psi}-2h\psi\right)=0, \tag{28}$$

$$\left(B_3+2B_4\right)\eta g\psi+2C_1\left(\kappa\psi+4gh\psi-g\dot{\psi}-\psi\dot{g}\right)+2\psi^3\left(2D_2-D_1-D_3\right)=0, \tag{29}$$

$$\begin{aligned}&B_3\left(\eta\left(h\psi-\dot{\psi}\right)-\psi\dot{\eta}\right)-2B_4\left(\eta\left(-h\psi-\dot{\psi}\right)-\psi\dot{\eta}\right)\\&+C_1\left(4h^2\psi+2\psi\dot{h}-\ddot{\psi}\right)+D_6\eta^2\psi=0,\end{aligned} \tag{30}$$

$$\begin{aligned}&B_3\left(\dot{g}+gh+2\kappa\right)\eta-2B_4\left(\dot{g}-gh\right)\eta+C_1\left(2\kappa h+4gh^2+2g\dot{h}-\ddot{g}\right)\\&+6h\psi^2\left(2D_2-D_1-D_3\right)+D_6\eta^2g-6F_3\eta\psi^2=0.\end{aligned} \tag{31}$$

We solve eq.(28) to find an expression for $\eta(t)$

$$\eta(t)=\left(\frac{2h\psi-\dot{\psi}}{\psi}\right)\left(\frac{C_1}{B_3-2B_4}\right), \tag{32}$$

replacing the above expression for $\eta(t)$ into eq. (30), leads to two sub-branches for the $h(t)$ function, either

$$h(t)=\frac{\dot{\psi}}{2\psi}\qquad\wedge\qquad h(t)=\frac{\dot{\psi}}{\psi}\left(\frac{C_1D_6}{3B_3^2-8B_3B_4+B_4^2+2C_1D_6}\right). \tag{33}$$

Using the simplest form of $h(t)$ function, then eq. (29) leads to

$$-2\left(D_1-2D_2+D_3\right)\psi^3+2C_1\left(\psi\left(\kappa-\dot{g}\right)+g\dot{\psi}\right)=0, \tag{34}$$

which allows to solve $g(t)$ in terms of the $\psi(t)$

$$g(t)=\psi(t)\left(g_0+\int_1^t\left(\frac{\kappa}{\psi(\tau)}-\psi(\tau)\left(\frac{D_1-2D_2+D_3}{C_1}\right)\right)\mathrm{d}\tau\right), \tag{35}$$

where g_0 is an integration constant. The above expression also solves eq. (31). Then, eq. (27) becomes an integro-differential equation of first order

$$\begin{aligned}&\dot{\psi}\left(g_0+\int_1^t\left(\frac{\kappa}{\psi(\tau)}-\psi(\tau)\left(\frac{D_1-2D_2+D_3}{C_1}\right)\right)\mathrm{d}\tau\right)\left(\frac{3B_3-2B_4}{2}\right)\\&-\psi^2\frac{\left(B_3-2B_4\right)\left(D_1-2D_2+D_3\right)+2C_1F_3}{C_1}+\kappa\left(3B_3-2B_4\right)=0.\end{aligned} \tag{36}$$

The above equation can be solved for the special case $\kappa = 0$, with the variable change $\psi(t) = \dot{\phi}(t)$, then eq. (36) turns into a second order differential equation for $\phi(t)$

$$\ddot{\phi}\left(g_0 - \phi\alpha\right)\beta - \dot{\phi}^2\gamma = 0, \tag{37}$$

where

$$\alpha = \left(\frac{D_1 - 2D_2 + D_3}{C_1}\right), \qquad \beta = \left(\frac{3B_3 - 2B_4}{2}\right), \tag{38}$$

$$\gamma = \frac{\left(B_3 - 2B_4\right)\left(D_1 - 2D_2 + D_3\right) + 2C_1F_3}{C_1}, \tag{39}$$

whose solution

$$\phi(t) = \frac{g_0}{\alpha} + \frac{\left(\phi_0\left(\alpha\beta + \gamma\right)\left(t + \phi_1\right)\right)^{\frac{\alpha\beta}{\alpha\beta+\gamma}}}{\alpha\beta}, \tag{40}$$

where ϕ_0 and ϕ_1 are integration constant. Using the above solution, it is straightforward to recover the original function

$$\psi(t) = \phi_0\left(\left(\alpha\beta + \gamma\right)\left(\phi_0 t + \phi_0\phi_1\right)\right)^{-\frac{\gamma}{\alpha\beta+\gamma}}. \tag{41}$$

From the above expression and using the relations in eqs. (32), (33) and (35) it is direct to find the rest of the affine functions

$$\eta(t) = 0, \tag{42}$$

$$h(t) = -\frac{\gamma}{2\left(\alpha\beta + \gamma\right)\left(t + \phi_1\right)}, \tag{43}$$

$$g(t) = \phi_0\left(\left(\alpha\beta + \gamma\right)\phi_0\left(t + \phi_1\right)\right)^{-\frac{\gamma}{\alpha\beta+\gamma}}\left(g_0 - \frac{\left(\left(\alpha\beta + \gamma\right)\phi_0\left(t + \phi_1\right)\right)^{\frac{\alpha\beta}{\alpha\beta+\gamma}}}{\beta}\right) \tag{44}$$

3.2 Second branch

The second branch, imposes the $g(t) = 0$ restriction, which leads to a reduced field equations

$$2\kappa B_3\psi - 2F_3\psi^3 = 0, \tag{45}$$

$$g = 0, \tag{46}$$

$$2\kappa C_1\psi - 2\psi^3\left(D_1 - 2D_2 + D_3\right) = 0 \tag{47}$$

$$\kappa\left(4h^2C_1 + h\eta\left(B_3 + 2B_4\right) + D_6\eta^2 + 2C_1\dot{h} - \dot{\eta}\left(B_3 - 2B_4\right)\right) = 0, \tag{48}$$

$$2h\left(\kappa C_1 - 3\left(D_1 - 2D_2 + D_3\right)\psi^2\right) + 2\eta\left(\kappa B_3 - 3F_3\psi^2\right) = 0. \tag{49}$$

From eq. (45) its possible to find an expression for $\psi(t)$

$$\psi(t) = \pm\sqrt{\frac{\kappa B_3}{F_3}}, \tag{50}$$

then, replacing the above algebraic expression in eq.(49) leads to a relation between $h(t)$ and $\eta(t)$ as follow

$$h(t) = -\frac{2B_3F_3\eta}{3B_3\left(D_1 - 2D_2 + D_3\right) - C_1F_3}. \tag{51}$$

Because of the above relation between the functions, eqs. (47) becomes a constraint equation for the coupling constant

$$\kappa\left(B_3\left(D_1 - 2D_2 + D_3\right) - C_1F_3\right) = 0 \rightarrow D_1 = \frac{2B_3D_2 - B_3D_3 + C_1F_3}{B_3}. \tag{52}$$

The constraint for the coupling constants, turns eq. (48) into a first order differential equation for $\eta(t)$

$$\dot{\eta}C_1\kappa\left(2B_4 - 3B_3\right) + \eta^2\left(3B_3^2 - 2B_3B_4 + C_1D_6\right)\kappa = 0, \tag{53}$$

which can be solved by

$$\eta(t) = \frac{C_1\left(2B_4 - 3B_3\right)}{\left(3B_3 - 2B_4\right)\left(tB_3 + \eta_0C_1\right) + tC_1D_6}, \tag{54}$$

then, the other the $h(t)$ can be written as follow

$$h(t) = \frac{B_3\left(3B_3 - 2B_4\right)}{\left(3B_3 - 2B_4\right)\left(tB_3 + \eta_0C_1\right) + tC_1D_6} \tag{55}$$

It is worth to notice the above nontrivial solution it is valid only for the special case where $\kappa \neq 0$, this can bee seen directly from eqs. (45), (52) and (53).

If one, impose the constraint $\kappa = 0$, then eq.(45) tell us that $\psi(t) = 0$ which solves entirely the other equations. Therefore, the other functions remain unknown.

3.3 Third branch

The constraint $\psi(t) = 0$, impose a strong restriction to the systems of differential equations, reducing to

$$g\eta^2D_6 + 2B_4\eta\left(gh - \dot{g}\right) + B_3\eta\left(2\kappa + gh + \dot{g}\right) + C_1\left(2h\left(\kappa + 2gh\right) + 2g\dot{h} - \ddot{g}\right) = 0. \tag{56}$$

The above differential equation has three unknown functions of time $h(t)$, $g(t)$ and $\eta(t)$ which cannot be solved without further restriction, or by providing an ansatz for two functions.

3.4 Fourth branch

In the fourth branch, defined by $g(t) = 0$ and $\psi(t) = 0$, the field equations are reduced to just one algebraic equation

$$\kappa \left(hC_1 + B_3 \eta \right) = 0 \tag{57}$$

the above equation can be solved by setting $\kappa = 0$, then, the functions $h(t)$ and $\eta(t)$ are undetermined, or by the relation $\eta(h) = -h(t) \left(\frac{C_1}{B_3} \right)$, where $h(t)$ is an arbitrary function.

4 Analysis of the solutions

The only nontrivial solutions can be found in Sec. 3.1, thus, in the following we shall focus our analysis in that branch. In order to simplify our calculations, and without any lost of generality, we shall set the integrations constant to $\phi_0 \to 1$, $\phi_1 \to 0$ and $g_0 \to 0$. Hence, the affine functions are simplify up to

$$h(t) = \frac{\gamma}{2t \left(\alpha\beta + \gamma \right)} \qquad g(t) = -\alpha^2 \beta \left(t \left(\alpha\beta + \gamma \right) \right)^{\frac{2\alpha\beta}{\alpha\beta+\gamma} - 1}, \tag{58}$$

$$\psi(t) = \alpha\beta \left(t \left(\alpha\beta + \gamma \right) \right)^{-\frac{\gamma}{\alpha\beta+\gamma}} \qquad \eta(t) = 0. \tag{59}$$

4.1 Emergent metrics

As mentioned in Sec. 2 it is possible to obtain emergent metric structures in this purely affine model of gravity. The fist type of metric like tensor comes from the Ricci tensor. Using eqs.(58) and (12) we compute the Ricci tensor

$$\mathcal{R}_{tt} = -\frac{3\gamma \left(2\alpha\beta + 3\gamma \right)}{4t^2 \left(\alpha\beta + \gamma \right)^2}, \tag{60}$$

$$\mathcal{R}_{rr} = \Sigma \left(\alpha, \beta, \gamma \right) t^{2\left(\frac{-\gamma}{\alpha\beta+\gamma} \right)}, \tag{61}$$

where $\Sigma \left(\alpha, \beta, \gamma \right)$ is defined as

$$\Sigma \left(\alpha, \beta, \gamma \right) = -\frac{\alpha^2 \beta \left(2\alpha\beta - 3\gamma \right) \left(\alpha\beta + \gamma \right)^{\frac{2\alpha\beta}{\alpha\beta+\gamma}}}{2 \left(\alpha\beta + \gamma \right)^2}. \tag{62}$$

It is worth to recall that the spatial component of the Ricci tensor plays an analogue role to the scale factor in the FLRW universe, thus

$$\Sigma \left(\alpha, \beta, \gamma \right) t^{2\left(\frac{-\gamma}{\alpha\beta+\gamma} \right)} \longleftrightarrow a^2(t). \tag{63}$$

Therefore, we can use eq.(61) to define notions of distance and angles, as long as is well behaved. For this metric, unlike the one in G.R., the metriciy condition is not satisfied, meaning $\nabla_\lambda \mathcal{R}_{\mu\nu}$ does not vanishes, and as a consequence of this, there are non-metricity effects $\mathcal{Q}^{\mathcal{R}}_{\mu\nu\lambda}$.

Additionally, it is also possible to compute Poplawski's metric like tensor which was defined in eq.(13) using eqs.(59), leading to

$$\mathcal{P}_{tt} = 0 \qquad\qquad \mathcal{P}_{rr} = \alpha^2\beta^2\left(t\left(\alpha\beta+\gamma\right)\right)^{-\frac{2\gamma}{\alpha\beta+\gamma}}, \tag{64}$$

however, unlike the Ricci tensor, the Poplawski metric like tensor, is a degenerate tensor which can not be inverted.

4.2 Matter emulation from torsion fields

Having a well defined affine descendent scale factor coming from the Ricci tensor, defined in eq.(61), is a power-law solution depending of the coupling constant α, β and γ, which are present due to the torsion field. In order to give a physical meaning, we can inquire the value of these parameters that possess a well-known behaviour in G.R., specifically in FLRW solutions.

Note that $\psi(t)$ has a similar form to the energy density expression in the standard model of cosmology

$$\psi(t) = \alpha\beta\left(t\left(\alpha\beta+\gamma\right)\right)^{-\frac{\gamma}{\alpha\beta+\gamma}} \longleftrightarrow \rho(t) = \rho_0 a^{-3(1+\omega)}, \tag{65}$$

meaning we could use the torsion tensor to have different types of affine scales factor. For example, consider the case with $\gamma = -\frac{\alpha\beta}{3}$, turns the affine scale factor

$$a_f(t) = a_0 t^{1/2}, \tag{66}$$

where a_0 its defined as

$$a_0 = \alpha^2\hat{\beta}, \tag{67}$$

with $\hat{\beta} = -\beta^3$, where β must be a negative number in order to have a real affine scale factor, which translate into an inequality for the coupling constants

$$3B_3 < 2B_4. \tag{68}$$

This type of solution allow us to emulate the radiation era.

In the same spirit, considering the case $\gamma = -\frac{2\alpha\beta}{5}$, leads to

$$a_f(t) = a_0 t^{2/3}, \tag{69}$$

where we defiend a_0 as

$$a_0 = \sqrt{-\alpha^{13/3}\beta^{10/3}}\left(\frac{2^{3/2}3^{2/3}}{5^{7/6}}\right), \tag{70}$$

which, in order to be real requires that $\beta < 0$, which is consistent with constraint written in eq. (68). This type of solution allow us to emulate a non-relativistic matter era.

If we want to have an accelerated expansion due to torsion effects, we need to consider

$$\ddot{a}_f > 0 \rightarrow \frac{\gamma\left(\alpha\beta+2\gamma\right)}{\left(\alpha\beta+\gamma\right)^2} t^{\frac{\alpha\beta}{\alpha\beta+\gamma}-3} > 0. \tag{71}$$

the above condition, ensures a positive acceleration. In order to have a steady growth acceleration, it is required that

$$\alpha\beta < -\frac{3}{2}\gamma, \tag{72}$$

this conditions ensures that the time evolution will be defined positive. Additionally, we have two branches coming from the numerator and denominator of eq. (71)

$$B1 : \alpha\beta > -2\gamma \qquad \gamma > 0, \tag{73}$$

$$B2 : \alpha\beta < -2\gamma \qquad \gamma < 0. \tag{74}$$

Using eq. (72) along with eqs. (73) and (74) allow us to define to intervals for the constant $\alpha\beta$, the former are $\alpha\beta \in \left[-2\gamma, -\frac{3}{2}\gamma\right]$, whereas the latter is given by $\alpha\beta < -2\gamma$.

5 Final remarks

In this work, we have analyzed some cosmological scenarios appearing in the context of the Polynomial Affine Gravity model with torsion effects, generalizing those considered in Refs. [42, 44]. Considering a non vanishing torsion tensor, yields an overdetermined system of differential equations, see eqs. (14) to (18), unlike the case without torsion effects, whose field equations are reduced to the *harmonic curvature* $\nabla_\alpha \mathcal{R}_{\beta\gamma}{}^{\alpha}{}_{\delta} = 0$, which through Bianchi's identity can be written as an anti symmetrized covariant derivative of the Ricci tensor $\nabla_{[\mu}\mathcal{R}_{\alpha]\beta} = 0$.

Without torsion effects, the field equation leads to one differential equation for two unknown functions (see eq.(18) with $\eta(t) = 0$), thus, the system can be solved parametrically, for example introducing an ansatz for $h(t)$ or $g(t)$. A priori, there is no physical reason to fix one of the functions, however, one could explore subspace of this torsion-free solutions, by studying integrability conditions such as $\nabla_\alpha \mathcal{R}_{\beta\gamma} = 0$. This, leads to a three differential equations, which can be solved analytically and provides with an exact solution for $h(t)$ and $g(t)$ functions. Even more, the Ricci tensor is well behaved and it can be used as a metric tensor. This case yields well-known space lie de-Sitter/Anti de-Sitter solutions, where the cosmological constant is an integration constant, see Ref. [44].

The introduction of torsion fields with the tensors $\mathcal{B}$ and $\mathcal{A}$, induce nontrivial effects in the dynamics of the system. Due to the presence of the torsion tensor, there are two emergent descendent metric structures coming from the irreducible fields of the affine connection. Specifically, one comes from the symmetric part of the connection, which allow us to define the covariant derivative ∇^Γ, and, the other one comes from the antisymmetric part of the affine connection, defined by the contraction of the product of two torsion tensors.

Keeping in mind in the standard FLRW model of the universe, there are two fundamental objects, the metric tensor $g_{\mu\nu}$, and the energy-momentum tensor

$T_{\mu\nu}$, we could argue that in Polynomial Affine Gravity, the Ricci tensor $\mathcal{R}_{\mu\nu}$ can act as $g_{\mu\nu}$, whereas $\mathcal{P}_{\mu\nu}$ emulate a $T_{\mu\nu}$. In that sense, the Ricci tensor provide us an affine descendent scale factor, while the Poplawski like metric tensor emulate different types of matter effects,

Interestingly, the system of field equations eqs. (14) to (18) present four different branches of solution, of which only one can be solved exactly (non parametrically). This solution, has a nontrivial traceless part of the torsion $\psi(t)$, that resembles the energy density from General Relativity. Because of that, we considered different configurations of α, β and γ parameters, and analyze their effects in the metric tensor coming from the Ricci. For specific values of those parameters, we were able to emulate the radiation era and the non-relativistic matter era. Additionally, we were able to define constraints for those parameters in order to secure an accelerated expansion, allowing us to emulate dark matter effects.

It has been sen that, the introduction of torsion effects in this affine model allow us to emulate matter effects in the emergent metric structure coming from the Ricci tensor. In this formulation, the three major eras, are due to the presence of torsion and not an energy-momentum tensor.

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